

Mazen Mohamed Hamdy

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OBJECTIVE

Information Engineering & Technology student at GUC with proficiency in Java, Python, Docker containerization, and networking fundamentals (TCP/IP, routing). Hands-on experience from homelab setups and hardware systems. Seeking a network engineering internship in Egypt to apply troubleshooting skills, contribute to infrastructure monitoring, and grow in enterprise networking.

EDUCATION

German University in Cairo

- Bachelor of Engineering, Dept. of Information Engineering and Technology
- Expected Graduation: 2029
- Current GPA: 1.65 (A-)
- Relevant Coursework: Digital Logic Design, Communication Networks, Java OOP, Data Structures & Algorithms.

TECHNICAL SKILLS

Languages: Java, Python

Tools & Software: Docker, Docker Compose, VirtualBox, WireGuard, nginx Proxy Manager, PSpice

Concepts: TCP/IP fundamentals, Routing and Switching basics, Subnetting, Network Troubleshooting, VPN fundamentals, Container Networking, Linux Server Administration, Java OOP, Data Structures, Algorithms, Boolean Logic Optimization

ENGINEERING PROJECTS

Hardware Encryption/Decryption System 2025

- Designed and prototyped a physical hardware system on a breadboard to encrypt and decrypt binary data.
- Applied fundamental logic gates and Boolean algebra to ensure reliable data processing.

OOP Java Game Engine 2026

- Engineered a modular game engine in Java with advanced OOP design and layered architecture.
- Developed features such as energy management, role-based mechanics, and event-driven cell interactions.
- Validated functionality through comprehensive unit testing and exception handling strategies.

Java Socket Chat Application 2026

- Built TCP client-server chat app in Java across multiple devices.
- Implemented bidirectional communication between client and server across different devices over a local network.
- Analyzed TCP packets with Wireshark (source/dest IP, packet flow).

Self-Hosted Homelab Infrastructure 2024

- Deployed containerized services using Docker and Docker Compose.
- Configured WireGuard VPN and nginx reverse proxy for secure access.
- Deployed Docker Compose media server infrastructure with automated download management.

EXTRA CURRICULAR

IGCSE Chemistry Academic Assistant & Technical Lead 2023-2026

- Supported chemistry teacher with IGCSE Grade 10 academic sessions, student concept clarification, and classroom management.
- Designed and deployed studio technical infrastructure for educational content production, including recording hardware, OBS setup, and supporting software workflow.

LANGUAGES

Arabic (Native), English (Fluent), German (A2 level).